

HOMEWORK - 2

(Axler)

Q. 2.A # 10

- 10 Prove or give a counterexample: If $v_1, v_2, \dots, v_m$ is a linearly independent list of vectors in V and $\lambda \in F$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, \dots, \lambda v_m$ is linearly independent.

Given that $\{v_1, v_2, \dots, v_n\} \in V$ is linearly independent,

it means:

$$\left. \begin{aligned} a_1 v_1 + a_2 v_2 + \dots + a_n v_n &= 0 \\ \text{where, } a_1 &= a_2 = \dots = a_n = 0 \end{aligned} \right\} \text{--- } \textcircled{1}$$

Let $\{\lambda v_1, \lambda v_2, \dots, \lambda v_n\}$ not be linearly independent

This means there exists $a_i \in \{a_1, a_2, a_3, \dots, a_n\} \in F$ s.t.

$a_i \neq 0$ and,

$$a_1(\lambda v_1) + a_2(\lambda v_2) + \dots + a_n(\lambda v_n) = 0$$

$$\lambda(a_1 v_1 + a_2 v_2 + \dots + a_n v_n) = 0$$

Since $\lambda \neq 0$,

$$(a_1 v_1 + a_2 v_2 + \dots + a_n v_n) = 0$$

$$\Rightarrow (a_1 v_1 + a_2 v_2 + \dots + a_i v_i + \dots + a_n v_n) = 0$$

and $a_i \neq 0$.

But this contradicts $\textcircled{1}$, $\{v_1, v_2, \dots, v_n\}$ is linearly independent.

$\therefore$ Our assumption must be false.

If $\{v_1, v_2, \dots, v_n\}$ are linearly independent, $\{\lambda v_1, \lambda v_2, \dots, \lambda v_n\}$ is also linearly independent.

Q 2.A # 16

16 Explain why no list of four polynomials spans $\mathcal{P}_4(F)$.

$\mathcal{P}_4(F)$ denotes the set of all polynomials with coefficients in F and degree at most 4. Let $p \in \mathcal{P}_4(F)$

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \quad \text{--- (1)}$$

where, $a_0, a_1, a_2, a_3, a_4 \in F$

The span of $\mathcal{P}_4(F) = \text{span}(1, x, x^2, x^3, x^4)$

Hence, $\{1, x, x^2, x^3, x^4\}$ is linearly independent

↳ Assume $\{1, x, x^2, x^3, x^4\}$ is lin. dep.

$$\therefore k_1(1) + k_2(x) + \dots + k_5(x^4) = 0 = p_0(x)$$

$$\exists k_i : k_i \in \{k_1, k_2, k_3, k_4, k_5\} \wedge k_i \neq 0$$

a polynomial of degree 4 has at most 4 roots
 $\therefore p_0(x) = 0$ for at most 4 values of x

Hence, $p_0(x) \neq 0 \quad \forall x \in F$. Our assumption was false, $\{1, x, x^2, x^3, x^4\}$ is lin. indep.

$\therefore B = \{1, x, x^2, x^3, x^4\}$ is the basis of $\mathcal{P}_4(F)$.

Since there are 5 elements in B :

$$\dim(\mathcal{P}_4(F)) = 5.$$

In order to span a set with dimension 5, the size of the spanning set must be ≥ 5 .

$\therefore$ A spanning set of 4 polynomials can never span $\mathcal{P}_4(F)$ since there need to be a minimum of 5 elements in a spanning set.

Q. 2.B #6

- 6 Prove or give a counterexample: If p_0, p_1, p_2, p_3 is a list in $\mathcal{P}_3(F)$ such that none of the polynomials p_0, p_1, p_2, p_3 has degree 2, then p_0, p_1, p_2, p_3 is not a basis of $\mathcal{P}_3(F)$.

A basis is a set of n linearly independent polynomials in $\mathcal{P}_{n-1}(F)$.

$\therefore$ for $\mathcal{P}_3(F)$, its basis would have:

$\Rightarrow$ 4 elements

$\Rightarrow$ all linearly independent

Counterexample:

$$\text{let } p_0 = 1,$$

$$p_1 = x$$

$$p_2 = x^3$$

$$p_3 = x^2 + x^3$$

Linear independence:

$$a + bx + cx^3 + d(x^2 + x^3) = 0 \quad (a, b, c, d \in F) \quad \text{--- } \textcircled{1}$$

$$a + bx + dx^2 + (c+d)x^3 = 0$$

Since we know that set $\{1, x, x^2, x^3\}$ is linearly independent (can be proven similar to how I did in 2A #16)

The coefficients must all be zero.

$$\therefore a = 0, \quad b = 0, \quad d = 0, \quad c + d = 0$$

$$\text{since } d = 0, \quad c + 0 = 0$$

$$c = 0$$

Hence only a trivial solution exists for eq. $\textcircled{1}$.

$\therefore p_0, p_1, p_2, p_3$ are linearly independent.

We now have a set of 4 linearly independent polynomials. $\therefore$ It is a basis. Counterexample found!

Q. 2.B # 8

- 8 Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U .

Counterexample:

$$\text{let } V = \mathbb{R}^4, \quad v_1 = e_1, \quad v_2 = e_2, \quad v_3 = e_3, \quad v_4 = e_4$$

$$\text{let } U = \text{span}(e_1, e_2, e_3 + e_4)$$

↳ U is a subset of V :

① zero: $0e_1 + 0e_2 + 0(e_3 + e_4) = 0 \in U$

② closed under addition:

$$\text{let } u = ae_1 + be_2 + c(e_3 + e_4) \quad (a, a', b, b', c, c' \in F)$$

$$u' = a'e_1 + b'e_2 + c'(e_3 + e_4)$$

$$\text{then, } u + u' = (a+a')e_1 + (b+b')e_2 + (c+c')(e_3 + e_4) \in U$$

③ closed under scalar multiplication:

let $\lambda \in F$ and u be as defined above.

$$\lambda u = \lambda(ae_1 + be_2 + c(e_3 + e_4))$$

$$= \lambda ae_1 + \lambda be_2 + \lambda c(e_3 + e_4) \in U$$

$v_1, v_2 \in U$ since,

$$v_1 = e_1 + 0e_2 + 0(e_3 + e_4)$$

$$v_2 = 0e_1 + e_2 + 0(e_3 + e_4)$$

But $v_3, v_4 \notin U$ since,

coefficient c will need to be 1 to include e_3 term but

simultaneously be 0 to not include e_4 to get v_3 which is impossible. Similarly for v_4 .

However,

$U \neq \text{span}(e_1, e_2)$ since,

$$e_3 + e_4 \notin \text{span}(e_1, e_2)$$

$\therefore v_1$ and v_2 don't span U , hence are not a basis for U . Counterexample found!

Q. 2.C # 4

- 4 (a) Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p''(6) = 0\}$. Find a basis of U .
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{R})$.
(c) Find a subspace W of $\mathcal{P}_4(\mathbb{R})$ such that $\mathcal{P}_4(\mathbb{R}) = U \oplus W$.

a) $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p''(6) = 0\}$
let $p_0 \in U$ and $a, b, c, d, e \in \mathbb{R}$

given constraint $p''(6) = 0$, we have,

$$p_0(x) = a + bx + cx^2 + dx^3 + ex^4 \quad (\because U \subseteq \mathcal{P}_4(\mathbb{R})) \quad \text{--- ①}$$

$$p_0'(x) = b + 2cx + 3dx^2 + 4ex^3$$

$$p_0''(x) = 2c + 6dx + 12ex^2$$

$$\hookrightarrow p_0''(6) = 2c + 36d + 432e = 0$$

$$\Rightarrow 2c + 36d + 432e = 0$$

$$c + 18d + 216e = 0$$

$$c = -18d - 216e$$

substituting in ①,

$$\begin{aligned} p_0(x) &= a + bx + (-18d - 216e)x^2 + dx^3 + ex^4 \\ &= a + bx + d(x^3 - 18x^2) + e(x^4 - 216x^2) \end{aligned}$$

$\therefore$ Every $p \in U$ can be written as a linear combination of,
 $1, x, (x^3 - 18x^2), (x^4 - 216x^2)$

$\therefore$ Set $A = \{1, x, (x^3 - 18x^2), (x^4 - 216x^2)\}$ is a spanning set for U .

consider,

$$a + bx + d(x^3 - 18x^2) + e(x^4 - 18x^2) = 0$$

$$a + bx + dx^3 - 18dx^2 + \underline{ex^4} - 18ex^2 = 0$$

ex^4 is the term of highest degree, but the RHS has no term x^4 , $\therefore$ coefficient $e = 0$.

$$\Rightarrow a + bx + dx^3 - 18dx^2 = 0$$

similarly, dx^3 is highest degree term, $\therefore d = 0$

$\Rightarrow a + bx = 0$, using same logic for bx , $b = 0$ and the $a = 0$.

$$\therefore a = b = c = d = 0$$

Hence, set A's elements are linearly independent

Since A spans U and is linearly independent, it is a basis.

$$\therefore \text{Basis of } U = \{1, x, x^3 - 18x^2, x^4 - 216x^2\}$$

b) $\dim(P_4(\mathbb{R})) = 5$

hence, to extend basis of U we need one more linearly independent element $\notin U$.

Any polynomial not satisfying $p''(6) = 0$ should work.

say $\underline{x^2}$,
 $(x^2)'' = (2x)' = 2 \neq 0 \notin U$.

Assume x^2 can be written as:

$$x^2 = a + bx + c(x^3 - 18x^2) + d(x^4 - 216x^2)$$

$$x^2 = a + bx + (-18c - 216d)x^2 + cx^3 + dx^4$$

$$\therefore a = b = c = d = 0 \quad (\because \text{no term of } 1, x, x^3, x^4 \text{ in LHS})$$

But if $c = d = 0$, then $-18c - 216d = 0 \neq 1$
 which is a contradiction. $\therefore$ Our assumption is false.

x^2 cannot be written in terms of elements of A.

$\therefore$ set $\{1, x, x^2, x^3 - 18x^2, x^4 - 216x^2\}$ is lin. indep.

$$\therefore \{1, x, x^2, x^3 - 18x^2, x^4 - 216x^2\} \text{ is an extended basis of } P_4(\mathbb{R})$$

c) from (a), $\dim(U) = 4$

we know $\dim(P_4(\mathbb{R})) = 5$

given $U \oplus W = P_4(\mathbb{R})$,

$$\dim(P_4(\mathbb{R})) = \dim(U + W)$$

$$= \dim(U) + \dim(W) - \dim(U \cap W)$$

$$5 = 4 + \dim(W) - 0 \quad (\because U \cap W = \{0\})$$

$$\therefore \dim(W) = 1$$

Hence, W needs to be a 1 dimensional space not in U

$$\therefore W = \text{span}\{x^2\} \quad (x^2 \notin U \text{ from (b)})$$

Q. 2. C # 12

- 12 Suppose that U and W are subspaces of $\mathbb{R}^8$ such that $\dim U = 3$, $\dim W = 5$, and $U + W = \mathbb{R}^8$. Prove that $\mathbb{R}^8 = U \oplus W$.

$U \oplus W = \mathbb{R}^8$ means $U + W = \mathbb{R}^8$ and $U \cap W = \{0\}$
for finite dimensional subspaces U and W ,
 $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$
substituting given values,

$$\begin{aligned}\Rightarrow \dim(\mathbb{R}^8) &= 8 = 3 + 5 - \dim(U \cap W) \\ \Rightarrow 8 &= 8 - \dim(U \cap W) \\ \Rightarrow 0 &= \dim(U \cap W)\end{aligned}$$

Hence, $U \cap W$ must be $\{0\}$.
Since $U + W = \mathbb{R}^8$ (given)
 $\therefore$ and $U \cap W = \{0\}$
 $U \oplus W = \mathbb{R}^8$ proven!

Q. 2. C # 15

- 15 Suppose V is finite-dimensional and V_1, V_2, V_3 are subspaces of V with $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Since, $V_1, V_2, V_3 \subseteq V$,

$$\dim(V_1 + V_2) \leq \dim V$$

$$\Rightarrow \dim(V_1 \cap V_2) = \dim V_1 + \dim V_2 - \dim(V_1 + V_2) \geq \dim V_1 + \dim V_2 - \dim V \quad \text{--- ①}$$

also,

$$(V_1 \cap V_2) \cap V_3 = V_1 \cap V_2 \cap V_3$$

$$\text{let } A = V_1 \cap V_2$$

$$\therefore \dim(A \cap V_3) = \dim A + \dim V_3 - \dim(A + V_3)$$

$$\therefore A + V_3 \subseteq V, \quad \dim(A \cap V_3) \leq \dim(V)$$

$$\begin{aligned}\therefore \dim(V_1 \cap V_2 \cap V_3) &= \dim(V_1 \cap V_2) + \dim(V_3) - \dim(V_1 \cap V_2 \cap V_3) \\ &\geq \dim(V_1 \cap V_2) + \dim(V_3) - \dim V\end{aligned}$$

using ①,

$$\begin{aligned}\dim(V_1 \cap V_2 \cap V_3) &\geq (\dim V_1 + \dim V_2 - \dim V) + \dim V_3 - \dim V \\ &= \dim V_1 + \dim V_2 + \dim V_3 - 2 \dim V \quad \text{--- ②}\end{aligned}$$

$$\text{given, } \dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$$

$$\text{from ②,} \\ \dim (V_1 \cap V_2 \cap V_3) > 0$$

this implies that $V_1 \cap V_2 \cap V_3 \neq \{0\}$!

Q. 3. A # 3

- 3 Suppose that $T \in \mathcal{L}(F^n, F^m)$. Show that there exist scalars $A_{j,k} \in F$ for $j = 1, \dots, m$ and $k = 1, \dots, n$ such that

$$T(x_1, \dots, x_n) = (A_{1,1}x_1 + \dots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \dots + A_{m,n}x_n)$$

for every $(x_1, \dots, x_n) \in F^n$.

Let $\{e_1, e_2, \dots, e_n\}$ be the standard basis for F^n s.t.
 $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$... $e_n = (0, 0, \dots, 1)$

$$\therefore \text{ for any } x = (x_1, x_2, x_3, \dots, x_n) \in F^n, \\ x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n \quad \text{--- ①}$$

$$\text{Every } T(e_k) \in F^m \quad (k \in \{1, 2, \dots, n\})$$

$$\therefore T(e_k) = (A_{1k}, A_{2k}, \dots, A_{mk}) \quad \rightarrow A_{jk} \in F, j \in \{1, \dots, m\}$$

For an arbitrary element $x \in F^n$,

using ① and
additivity of linear
maps.

$$\begin{aligned} T(x) &= T(x_1 e_1) + T(x_2 e_2) + \dots + T(x_n e_n) \\ &= x_1 (A_{11}, \dots, A_{m1}) + \dots + x_n (A_{1n}, \dots, A_{mn}) \\ &= (x_1 A_{11}, \dots, x_n A_{m1}) + \dots + (x_n A_{1n}, \dots, x_n A_{mn}) \\ &= (x_1 A_{11} + x_2 A_{21} + \dots + x_n A_{n1}, \dots, x_n A_{1n} + x_n A_{2n} + \dots + x_n A_{mn}) \end{aligned}$$

Since our choice of $x \in F^n$ was arbitrary (along with any other variables defined) we can generalize to,

for any $T \in \mathcal{L}(F^n, F^m)$ there exist scalars $A_{j,k} \in F$
such that,

$$T(x_1, \dots, x_n) = (A_{11}x_1 + \dots + A_{1n}x_n, \dots, A_{m1}x_1 + \dots + A_{mn}x_n)$$

proven!

Q. 3.A #5

5 Prove that $\mathcal{L}(V, W)$ is a vector space, as was asserted in 3.6.

If $\mathcal{L}(V, W)$ satisfies A1-A4 and S1-S4 then it is a vector space.

A1 & S3 : Associative

$$\begin{aligned} \text{Let } T_1, T_2, T_3 \in \mathcal{L}(V, W) \text{ and } v \in V \\ \Rightarrow ((T_1 + T_2) + T_3)(v) &= (T_1 + T_2)(v) + T_3(v) \\ &= (T_1(v) + T_2(v)) + T_3(v) \\ &= T_1(v) + (T_2(v) + T_3(v)) \quad (\text{Associativity of } W) \\ &= T_1(v) + (T_2 + T_3)(v) \\ &= (T_1 + (T_2 + T_3))(v) \end{aligned}$$

$\Rightarrow$ Let $a, b \in F$.

$$\begin{aligned} ((ab)T_1)(v) &= (ab)T_1(v) \\ &= a(bT_1(v)) \quad (\text{S3 over } W) \\ &= a((bT_1)(v)) \end{aligned}$$

A2: Zero :

$$\begin{aligned} \text{Let } 0_V \in \mathcal{L}(V, W) \text{ s.t. } 0_V + T = T \quad \forall T : T \in \mathcal{L}(V, W) \\ \therefore 0_V(v) = 0_W \quad (0_W \in W, \text{ zero of } W) \\ (T + 0_V)(v) &= T(v) + 0_V(v) \\ &= T(v) + 0_W \\ &= T(v) \quad (\text{zero of } W) \end{aligned}$$

A3: Additive Inverse:

$\forall T : T \in \mathcal{L}(V, W)$, let there be a $-T$ s.t. $T + (-T) = 0_V$

$\therefore -T(v) \Rightarrow$ additive inverse of $T(v)$ in W .

$$\begin{aligned} (T + (-T))(v) &= T(v) + (-T)(v) \\ &= 0_W \quad (\text{Additive inverse in } W) \end{aligned}$$

A4: Commutativity (Addition):

$$\begin{aligned} (T_1 + T_2)(v) &= T_1(v) + T_2(v) \\ &= T_2(v) + T_1(v) \quad (\text{commutativity in } W) \\ &= (T_2 + T_1)(v) \end{aligned}$$

S1 & S2: Distributive :

$$\begin{aligned} ((a+b)T)(v) &= (a+b)T(v) \\ &= aT(v) + bT(v) \quad (\text{distributivity of } W) \\ &= (aT + bT)(v) \end{aligned}$$

$$\begin{aligned} (a(T_1 + T_2))(v) &= a(T_1 + T_2)(v) = a(T_1(v) + T_2(v)) \\ &\stackrel{\text{distributivity of } W}{=} aT_1(v) + aT_2(v) = (aT_1 + aT_2)(v) \end{aligned}$$

S4: Multiplicative Identity:

$$\begin{aligned}(1T)(v) &= 1T(v) \\ &= T(v) \quad (\text{multiplicative identity of } w)\end{aligned}$$

$\therefore$ Since, all required axioms have been satisfied,
 $L(V, w)$ is a vector space!

Q. 3.6 # 3

3 Suppose $v_1, \dots, v_m$ is a list of vectors in V . Define $T \in \mathcal{L}(F^m, V)$ by

$$T(z_1, \dots, z_m) = z_1 v_1 + \dots + z_m v_m.$$

- (a) What property of T corresponds to $v_1, \dots, v_m$ spanning V ?
- (b) What property of T corresponds to the list $v_1, \dots, v_m$ being linearly independent?

a) T essentially takes an m tuple in F^m , and returns a linear combination of vectors $v_1, \dots, v_m$.

Hence, $\text{range}(T) = \text{span}(v_1, \dots, v_m)$ — ①

Given that $\text{span}(v_1, \dots, v_m) = V$, and ①, implies,

$$\text{range}(T) = V. \quad \text{--- ②}$$

Surjectivity is defined as if $T: V \rightarrow W$, then T is surjective if $\text{range of } T = W$.

Hence, from ②, corresponding property is surjectivity.

b) if $v_1, \dots, v_m$ is linearly independent,

$$z_1 v_1 + \dots + z_m v_m = 0 \quad \text{where}$$

$$z_1 = z_2 = \dots = z_m = 0$$

This is equivalent of,

$$T(z_1, \dots, z_m) = 0 \rightarrow (z_1, \dots, z_m) = 0$$

which means only one vector in F^m maps to one output (here 0).

$\therefore$ corresponding property is one-to-one

Q. 3.B #5

5 Give an example of $T \in \mathcal{L}(\mathbb{R}^4)$ such that $\text{range } T = \text{null } T$.

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^4 \quad (\text{given})$$

Using rank-nullity theorem,

(or as mentioned in Axler as Fundamental Theorem of Linear Maps)

$$\dim(\mathbb{R}^4) = \dim \text{null } T + \dim \text{range } T$$

— ①

here, domain and range are both $\subseteq \mathbb{R}^4$.

If they are equal,

$$\text{①} \Rightarrow 4 = k + k$$

$$(k = \dim \text{null } T = \dim \text{range } T)$$

$$k = 2 \rightarrow \text{2 dimensional space.}$$

$$\text{let } T(e_1) = e_3$$

$$T(e_3) = 0$$

$$T(e_2) = e_4$$

$$T(e_4) = 0$$

$$\text{--- ② } e_1, e_2, e_3, e_4 \in \mathbb{R}^4$$

here for an arbitrary vector,

$$x = ae_1 + be_2 + ce_3 + de_4 \quad (a, b, c, d \in \mathbb{F})$$

$$T(x) = aT(e_1) + bT(e_2) + cT(e_3) + dT(e_4)$$

$$= ae_3 + be_4 + 0 + 0$$

$$= ae_3 + be_4$$

— ③

$$\text{from ②, } \text{null } T = \text{span}(e_3, e_4)$$

$$\text{from ③, } \text{range } T = \text{span}(e_3, e_4)$$

} both of dim = 2!

$\therefore$ example of T :

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^4 \quad \text{s.t.} \quad \begin{aligned} T(e_1) &= e_3, & T(e_3) &= 0 \\ T(e_2) &= e_4, & T(e_4) &= 0 \end{aligned}$$

Q. 3.B #6 :

6 Prove that there does not exist $T \in \mathcal{L}(\mathbb{R}^5)$ such that $\text{range } T = \text{null } T$.

Assume there exists a $T \in \mathcal{L}(\mathbb{R}^5)$ s.t. $\text{range } T = \text{null } T$.

using rank-nullity,

$$\dim(\mathbb{R}^5) = \dim \text{null } T + \dim \text{range } T$$

$$5 = k + k$$

$$(\because \text{range } T = \text{null } T)$$

$$k = \frac{5}{2}$$

But k must be an integer $\because$ its a dimension. $\therefore$ contradiction!
Our assumption was false.